



KCED

Learning Augmented

24ADI003-MACHINE LEARNING

Unit III – CLASSIFICATION MODELS

Logistic Regression-Naive Bayes Classifiers-Decision Tree-K-Nearest Neighbors-Support Vector Machine – Evaluation metrics – AUC ROC- Class Imbalance – SMOTE – Cross-Validation Techniques-Ensemble Learning-Bagging- Random Forests-Boosting - AdaBoost -Gradient Boosting

NAIVE BAYES CLASSIFIER

In many real-world problems, we must **classify objects**:

- Is an email **Spam or Not Spam**?
- Will a student **Pass or Fail**?
- Is a transaction **Fraud or Genuine**?

Naive Bayes solves **classification problems** by using **probability theory**.

Instead of guessing directly, it asks:

“Given what I see, which class is most likely?”

Bayes' Theorem

- Bayes' theorem allows us to **reverse conditional probabilities**:

$$P(C | X) = \frac{P(X | C) P(C)}{P(X)}$$

Term	Meaning
$P(C)$	Prior probability (how frequent the class is)
$P(X)$	Evidence (same for all classes)

Since $P(X)$ is the same for all classes:

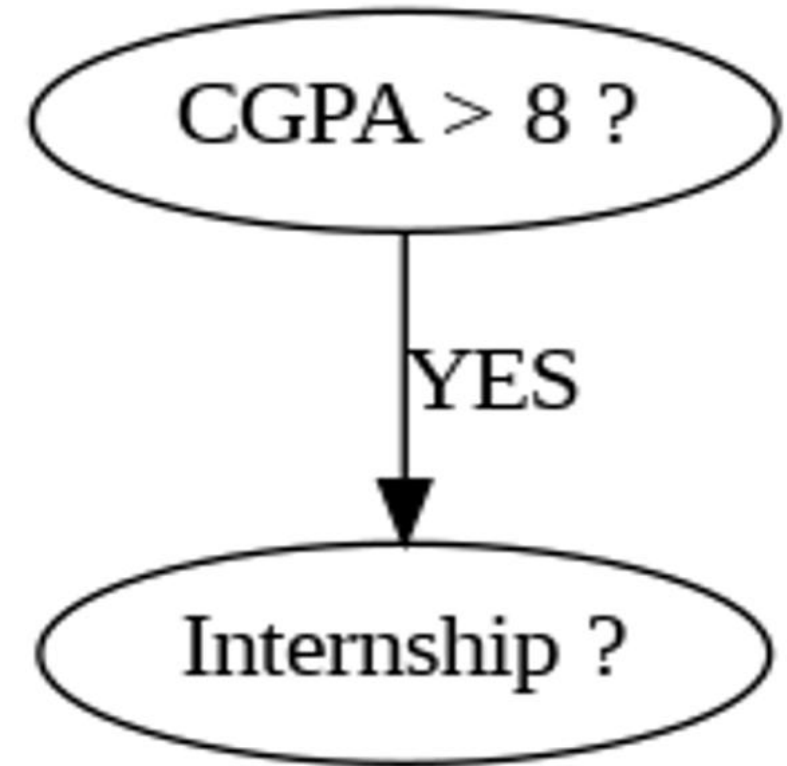
$$P(C | X) \propto P(X | C) P(C)$$

→ We **do NOT calculate** $P(X)$ in practice.

DECISION TREE CLASSIFIERS

- Decision Trees mimic **human decision making**.
- Example – Placement Prediction:

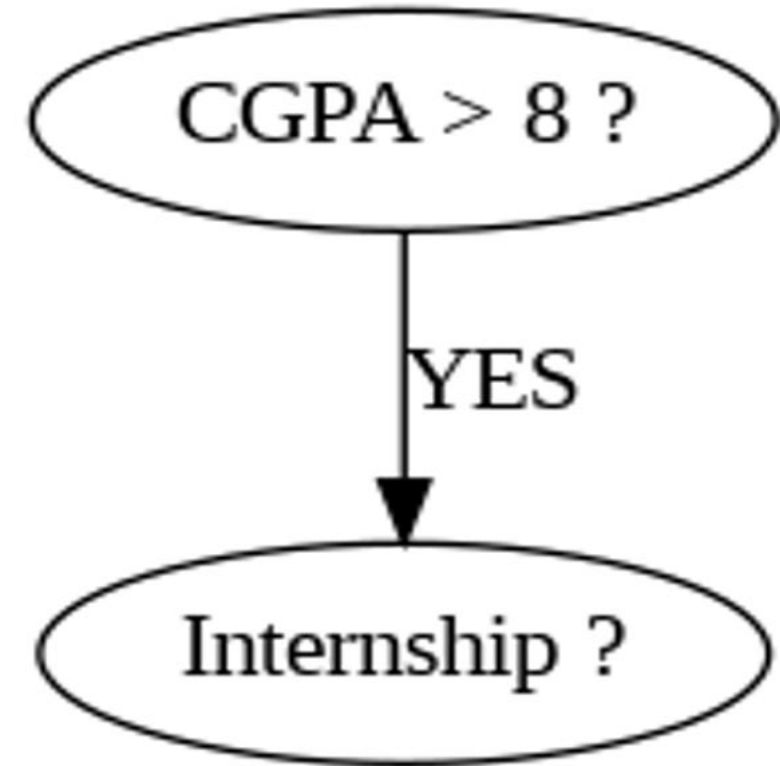
- So tree = **sequence of questions** → **prediction**
- Used in:
 - Medical diagnosis
 - Loan approval
 - Fraud detection
 - Student performance prediction



DECISION TREE CLASSIFIERS

- DECISION TREE TERMINOLOGY

Term	Meaning
Root Node	First question
Internal Node	Decision point
Leaf Node	Final output
Branch	Decision outcome
Depth	Length root → leaf
Parent/Child	Node relationship



DECISION TREE CLASSIFIERS

- **HOW DECISION TREES WORK – STEP-BY-STEP**

- Calculate impurity of dataset
- Try split using each feature
- Compute Information Gain
- Choose best split
- Repeat until pure
- Goal → **Make nodes pure**
- Pure node → all samples same class.

DECISION TREE CLASSIFIERS

- **NODE IMPURITY MEASURES**
- **ENTROPY**
- Entropy measures **uncertainty**.
- Formula:

$$H(S) = - \sum p_i \log_2(p_i)$$

- **Entropy Values**

Distribution	Entropy
100% Yes	0
50% Yes 50% No	1
80% Yes 20% No	0.72

- **Lower entropy = better split.**

DECISION TREE CLASSIFIERS

- **EXAMPLE**

- Dataset: – **ENTROPY**

Pass	Fail
6	4

Step 1 – Probabilities

- $p(\text{Pass}) = 6/10 = 0.6$
 $p(\text{Fail}) = 4/10 = 0.4$

Step 2 – Logs

- $\log_2(0.6) = -0.737$
 $\log_2(0.4) = -1.322$

Step 3 – Multiply

- $-0.6 \times (-0.737) = 0.442$
 $-0.4 \times (-1.322) = 0.529$

Step 4 – Add

- Entropy = **0.971**
- Interpretation → Dataset is mixed.

DECISION TREE CLASSIFIERS

- **EXAMPLE**

- Dataset: – **ENTROPY** Entropy = **0.971**

Pass	Fail
6	4

- **INFORMATION GAIN – FULL SOLUTION**

- Split by **Study Hours**

Group	Pass	Fail
Low	2	3
High	4	1

- **Step 1 – Parent Entropy**

- Already = 0.971

- **Step 2 – Entropy Low**

- $p = 2/5 = 0.4$
Entropy_{low} = 0.971

DECISION TREE CLASSIFIERS

- **INFORMATION GAIN – FULL SOLUTION**

- Split by **Study Hours**

Group	Pass	Fail
Low	2	3
High	4	1

- **Step 3 – Entropy High**

- $p = 4/5 = 0.8$

- Entropy_high =
 $= -0.8 \log_2 0.8 - 0.2 \log_2 0.2$
 $= \mathbf{0.722}$

- **Step 4 – Weighted Entropy**

- $= (5/10) \times 0.971 + (5/10) \times 0.722 = 0.485 + 0.361 = \mathbf{0.846}$

- **Step 5 – Information Gain**

- $IG = 0.971 - 0.846 = 0.125$

- **Small IG → Weak feature.**

- Students should compare IG for all features.

DECISION TREE CLASSIFIERS

- **GINI INDEX –**
- Formula:

$$Gini = 1 - \sum p_i^2$$

- Parent:

$$= 1 - (0.6^2 + 0.4^2) = 1 - (0.36 + 0.16) = 0.48$$

- Low Hours:

$$= 1 - (0.4^2 + 0.6^2) = 0.48$$

- High Hours:

$$= 1 - (0.8^2 + 0.2^2) = 0.32$$

- **Lower Gini = Better.**

DECISION TREE CLASSIFIERS

- PROBLEM STATEMENT-Placement Prediction DATASET

CGPA>8	Internship	Placed
Yes	High	Yes
Yes	Medium	Yes
Yes	Low	No
No	High	No
No	Medium	No

- **Step 1 – Parent Entropy**

- Yes = 2
No = 3

$$H = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5}$$

- = 0.971

DECISION TREE CLASSIFIERS

- PROBLEM STATEMENT-Placement Prediction DATASET

CGPA>8	Internship	Placed
Yes	High	Yes
Yes	Medium	Yes
Yes	Low	No
No	High	No
No	Medium	No

- Step 2 – Split on CGPA**

- CGPA=Yes → 2 Yes,1 No
CGPA=No → 0 Yes,2 No

- Entropy(CGPA Yes):

$$= -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.918$$

- Entropy(CGPA No) = 0

- Weighted entropy:

$$= \frac{3}{5} \times 0.918 + \frac{2}{5} \times 0 = 0.551$$

- IG(CGPA) = 0.971 – 0.551 = **0.42**

DECISION TREE CLASSIFIERS

- PROBLEM STATEMENT-Placement Prediction DATASET

Step 3 – Split on Internship

- Compute similarly $\rightarrow$ IG smaller.
- So **CGPA is best root.**



DECISION TREE CLASSIFIERS

OVERFITTING & PRUNING IN DECISION TREES

- **Overfitting** happens when a model learns the **training data too perfectly**, including noise and random patterns.

So it performs:

- Very high accuracy on training data
Poor accuracy on new/test data
- The model memorizes instead of learning.
- Suppose student placement data has 100 students.
- Model Result:

Dataset	Accuracy
Training	100%
Test	55%

- This means the tree memorized special cases like:

IF CGPA>8 AND Internship=Yes AND Name=Ravi → Placed

- Using **Name** is meaningless → Overfitting.

DECISION TREE CLASSIFIERS

OVERFITTING & PRUNING IN DECISION TREES



DECISION TREE CLASSIFIERS

OVERFITTING & PRUNING IN DECISION TREES

WHY OVERFITTING HAPPENS IN DECISION TREES

Decision Trees keep splitting until nodes are pure.

This can create:

- Very deep trees
- Too many branches
- Splits based on noise

Causes

1. Small dataset
2. Too many features
3. No pruning
4. Outliers
5. Imbalanced data

EFFECTS OF OVERFITTING

- Poor prediction on new data
- High variance model
- Complex tree hard to interpret
- Wasted computation

LEARNING
AUGMENTED

DECISION TREE CLASSIFIERS

OVERFITTING & PRUNING IN DECISION TREES

SOLUTIONS TO OVERFITTING

1. LIMIT TREE DEPTH

Depth

Depth = number of levels from root to leaf.

Example:

Depth 1 → Only 1 split

Depth 10 → Many splits

Why Limit Depth?

Deep tree learns noise.

So we stop tree early.

Example

Dataset of 50 students.

Depth=1 accuracy → 70%

Depth=5 accuracy → 85%

Depth=20 accuracy → 100% training but 60% test

Best depth ≈ 5

**Limiting depth reduces overfitting
Makes tree simple & interpretable**

DECISION TREE CLASSIFIERS

OVERFITTING & PRUNING IN DECISION TREES

SOLUTIONS TO OVERFITTING

2. MINIMUM SAMPLES PER LEAF

A **leaf node** should have enough data.
If leaf has only 1 sample → memorization.

Example

If tree splits until each leaf has 1 student:

IF RollNo=24BAD042 → Pass

Not useful.

Solution

Set minimum samples.

Example:

min_samples_leaf = 5

Means each leaf must have at least 5 records.

Effect

Reduces noise

Smooth predictions

Better generalization

EXAMPLE

Dataset = 100 students.

Without rule → 100 leaves

With min_samples_leaf=10 → only 10 leaves

Tree becomes stable.

DECISION TREE CLASSIFIERS

OVERFITTING & PRUNING IN DECISION TREES

SOLUTIONS TO OVERFITTING

3. PRUNING

Pruning = Cutting unnecessary branches.

Types of Pruning

1. Pre-Pruning (Early Stopping)

Stop tree before it becomes too deep.

Conditions:

max_depth

min_samples_split

min_samples_leaf

min_information_gain

Example

Stop splitting if IG < 0.01

Ignore weak splits.

2. Post-Pruning (Reduced Error Pruning)

Build full tree first → then remove branches that don't improve accuracy.

Steps:

1. Build full tree

2. Check each branch

3. Remove if accuracy doesn't drop

Example

Before pruning:

CGPA > 8?
AND Internship = Yes?
AND Library > 3?

After pruning:

CGPA > 8?
AND Internship = Yes?

Simpler and better.

Cost-Complexity Pruning (CART)

Minimize:

Error + $\alpha \times$ Tree Size

Balances accuracy & simplicity.

DECISION TREE CLASSIFIERS

OVERFITTING & PRUNING IN DECISION TREES

SOLUTIONS TO OVERFITTING

4. CROSS-VALIDATION

Cross-validation tests model on multiple splits.

Common method → **k-fold cross-validation**

Steps

Split data into k parts

Train on k-1 parts

Test on remaining part

Repeat k times

Take average accuracy

Example (k=5)

Data = 100 students

Each fold = 20 students.

Accuracy results:

70%, 72%, 75%, 71%, 74%

Average = 72.4%

Choose model with best average accuracy.

Why Useful?

Detect overfitting

Choose best depth

Choose best parameters

EXAMPLE:

College predicts pass/fail.

Tree depth 15 → 100% training accuracy

Test accuracy 58%

After pruning depth=4:

Training accuracy 88%

Test accuracy 84%

Better generalization.

K-NEAREST NEIGHBORS (KNN)

- KNN works like **human reasoning**.

“People similar to you behave like you.”

- Example – Student prediction:
- If most nearby students passed → You will pass.
- So prediction depends on **neighbors**.
- **Why called Lazy Learner?**
- Because KNN:
- Does **no training**
Stores all data
Predicts only when test data comes

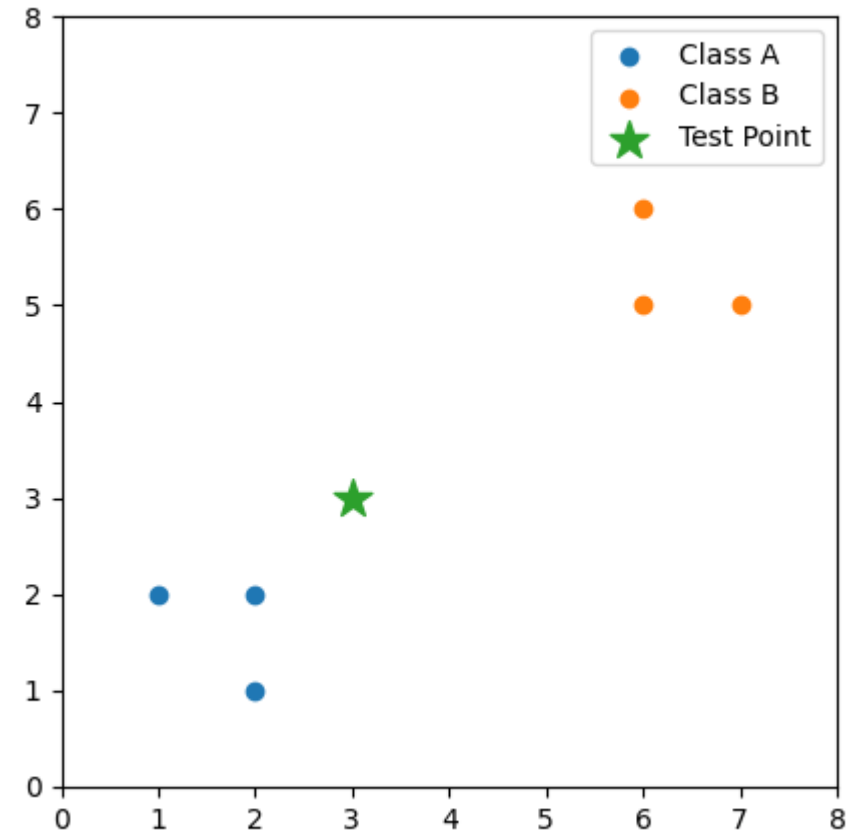
So memory heavy but simple.

- **Real-Life Examples**
- Movie recommendation
- Image recognition
- Spam detection
- Medical diagnosis
- Student performance prediction

K-NEAREST NEIGHBORS (KNN)

KNN – STEP BY STEP

- Suppose we want to predict Pass/Fail.
- Steps:
- Choose K (number of neighbors)
- Measure distance from test point to all training points
- Pick K nearest points
- Majority vote
- That class becomes prediction.



K-NEAREST NEIGHBORS (KNN)

DISTANCE METRICS

1. EUCLIDEAN DISTANCE

- Used for continuous data.

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- Example:
- (2,3) and (5,7)

$$\sqrt{(3)^2 + (4)^2} = 5$$

Straight line distance.

2. MANHATTAN DISTANCE

- City-block distance.

$$d = |x_1 - x_2| + |y_1 - y_2|$$

- Example:
- (2,3) → (5,7)

$$= 3 + 4 = 7$$

Move only horizontal/vertical.

3. COSINE SIMILARITY

Used for text data.

$$\cos \theta = \frac{A \cdot B}{|A| |B|}$$

Measures **angle**, not distance.

Used in:

- Document similarity
- Search engines
- Recommendation systems

K-NEAREST NEIGHBORS (KNN)

PROBLEM STATEMENT KNN

- Training Data

Math	Physics	Result
40	50	Fail
45	60	Fail
70	80	Pass
80	90	Pass

- Test Student = (60,70)
K = 3

K-NEAREST NEIGHBORS (KNN)

PROBLEM STATEMENT KNN

- Training Data

Math	Physics	Result
40	50	Fail
45	60	Fail
70	80	Pass
80	90	Pass

- Test Student = (60,70)
K = 3

Step 1 – Distance Calculation

Distance to (40,50)

$$\sqrt{(60 - 40)^2 + (70 - 50)^2}$$

$$= \sqrt{20^2 + 20^2} = \sqrt{800} = 28.28$$

Distance to (45,60)

$$\sqrt{15^2 + 10^2} = \sqrt{325} = 18.03$$

Distance to (70,80)

$$\sqrt{10^2 + 10^2} = \sqrt{200} = 14.14$$

Distance to (80,90)

$$\sqrt{20^2 + 20^2} = 28.28$$

K-NEAREST NEIGHBORS (KNN)

PROBLEM STATEMENT KNN

- Training Data

Math	Physics	Result
40	50	Fail
45	60	Fail
70	80	Pass
80	90	Pass

- Test Student = (60,70)
K = 3

Step 2 – Sort Distances

Distance	Result
14.14	Pass
18.03	Fail
28.28	Fail
28.28	Pass

- Nearest 3 → Pass, Fail, Fail

K-NEAREST NEIGHBORS (KNN)

PROBLEM STATEMENT KNN

Step 2 – Sort Distances

Distance	Result
14.14	Pass
18.03	Fail
28.28	Fail
28.28	Pass

• Step 3 – Majority Vote

- Fail = 2
- Pass = 1

- Prediction → **FAIL**

• Interpretation

- Student performance similar to failing students.
- So predicted Fail.

K-NEAREST NEIGHBORS (KNN)

KNN REGRESSION (Problem Statement)

- Predict salary.
- Nearest salaries:
- 30k, 32k, 40k
- Prediction = Average

$$(30 + 32 + 40)/3 = 34k$$

- KNN works for regression also.

EFFECT OF K VALUE

K Value	Result
Small K	Sensitive to noise
Large K	Too smooth

• Example

- Dataset has one wrong label.
- K=1 → Wrong prediction
K=5 → Correct prediction
- **Small K → Overfitting**
Large K → Underfitting

Best K Selection

Use Cross-Validation.

Try K=1 to 20 → Choose best accuracy

K-NEAREST NEIGHBORS (KNN)

NORMALIZATION IN KNN

- Very important!
- Because KNN uses distance.
- If one feature has large scale → dominates.

Example

Feature	Value
Height	170
Marks	80

Height dominates distance.

Solution

- **Min-Max Scaling**

$$x' = \frac{x - \min}{\max - \min}$$

- **Standardization**

$$z = \frac{x - \mu}{\sigma}$$

- Makes features comparable.

K-NEAREST NEIGHBORS (KNN)

PROBLEM STATEMENT

Points:

- (2,3) A
(3,4) A
(7,8) B
(8,9) B
- Test = (4,5), k=3

Step 2 – Sort

1.41 → A

2.82 → A

4.24 → B

Step 3 – Majority

A = 2

B = 1

Prediction → **A**

Step 1 – Distances

To (2,3)

$$\sqrt{(2)^2 + (2)^2} = 2.82$$

To (3,4)

$$\sqrt{1^2 + 1^2} = 1.41$$

To (7,8)

$$\sqrt{3^2 + 3^2} = 4.24$$

To (8,9)

$$\sqrt{4^2 + 4^2} = 5.65$$

K-NEAREST NEIGHBORS (KNN)

ADVANCED KNN IDEAS

Weighted KNN

- Closer neighbors get more importance.
- Weight = $1 / \text{distance}$.
- Example:
 - Distance 1 $\rightarrow$ Weight 1
 - Distance 5 $\rightarrow$ Weight 0.2
- More accurate prediction.

KD-Tree & Ball Tree

- Used to speed up KNN search.
- Important for big data.

Curse of Dimensionality

Too many features $\rightarrow$ distances become meaningless.

Example:

100 features $\rightarrow$ all points equally far.

Solution:

PCA

Feature selection

Remove irrelevant features

K-NEAREST NEIGHBORS (KNN)

KNN VS DECISION TREE

Feature	Decision Tree	KNN
Training	Fast	None
Prediction	Fast	Slow
Memory	Low	High
Interpretability	High	Low
Scaling Needed	No	Yes